CertiVar: Certified Likelihood Ratios for Variant-Effect Predictor Calls, Calibrated on Saturation Mutagenesis Data

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Abstract

Variant-effect predictors are a common source of evidence for missense variants of uncertain significance (VUS), and the Bayesian form of the ACMG/AMP clinical guidelines expresses evidence strength as a likelihood ratio. Gene-specific calibrations estimate this ratio from few labelled variants, as point estimates or resampling-based bounds that are not designed to remain valid once a threshold has been chosen from the same data. We present CertiVar, which turns a predictor score into loss-of-function (LoF) calls, functional calls or abstentions with two class-conditional conformal thresholds learned from saturation genome editing or deep mutational scanning data, keeping one variant per residue because substitutions at a residue are dependent. We prove that exact bounds on the likelihood ratio of each call type with respect to the assay outcome remain valid after a threshold is selected from a grid, by a Bonferroni correction or by fixed-sequence testing, and that per-class coverage after selection is at least one minus the largest grid level. Because every guarantee conditions on the assay class, none requires the share of LoF variants among the variants being called, provided their within-class score distribution matches that of the assayed variants; a mixture-consistency check tests a necessary condition for this on unlabelled targets. On 14,477 assayed missense variants in BRCA1, BRCA2, BAP1, RAD51C and MSH2, scored by AlphaMissense, REVEL and ESM-2, the lower of the two per-class coverages was 0.896–0.908 at a 0.90 target in all 15 gene–predictor pairs, with 35% abstention on average, and stayed at 0.89 or above when the calibration LoF share was varied from 5% to 50%; marginal conformal prediction calibrated at 5% covered LoF variants at only 0.11 in its worst pair. Against known population likelihood ratios, the certified bounds all held together in 99.7% of 1,500 resampled data sets (guaranteed: at least 90%), against 57.6% for percentile-bootstrap bounds. At the 5% level, the mixture check rejected 4.7% of random splits and every tested set of unassayed VUS outside the assayed residues. The certificates concern the assay, not pathogenicity; certified evidence was mostly supporting-level, required same-gene assay data, and held for 88.8% of bounds in held-out protein regions, below the nominal 95%. Code: `https://github.com/aaryavvatts-lab/certivar`

Keywords: variant effect prediction; conformal prediction; likelihood ratio; ACMG/AMP evidence calibration; multiplexed assays of variant effect; variants of uncertain significance; selective classification; multiple testing.

1 Introduction

In the five cancer-predisposition genes studied here, 81% of the 11,010 ClinVar missense records classified as (likely) pathogenic, (likely) benign or of uncertain significance are variants of uncertain significance (VUS; our count from the release of 26 October 2026 [1]), and for many of them computational predictors are among the few sources of evidence [2]. The guidelines of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology (ACMG/AMP) grade evidence qualitatively [3]; their Bayesian adaptation equates each strength with a likelihood ratio, the odds of pathogenicity (OddsPath) [4, 5]. Calibration workflows report this ratio as a point estimate or a resampling-based bound [2, 6, 7]. For a single gene, few labelled variants are available and the threshold that defines the evidence is chosen from the same data; these bounds are not designed to remain valid after that choice.

Two features of the data make the estimate fragile. First, the share of damaging variants differs between calibration sets and the variants being classified, and is unknown for the latter: in our genes, 45% of confidently labelled ClinVar missense variants are pathogenic, whereas 6.0% to 25.9% of the other assayed missense variants lose function in saturation genome editing (SGE) or deep mutational scanning (DMS)

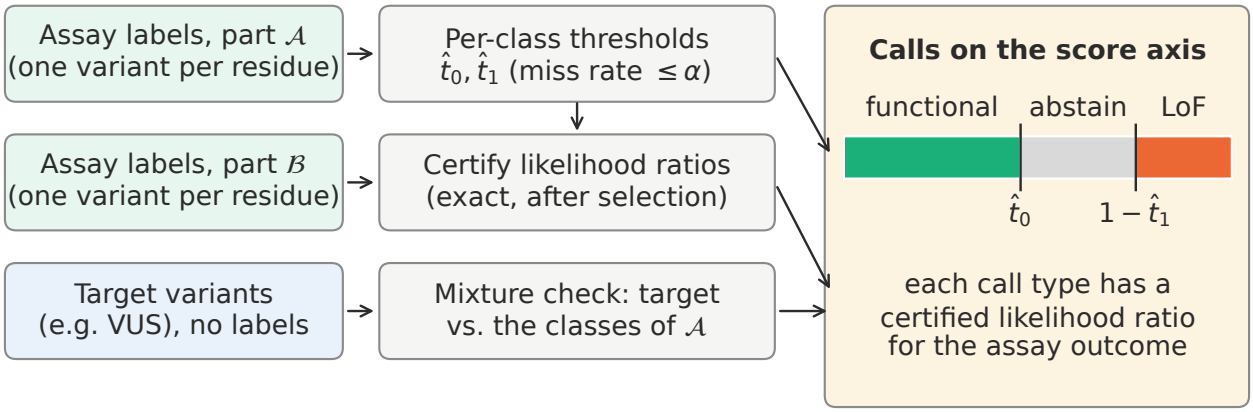


Figure 1: CertiVar. Per-class thresholds are set on part $\mathcal{A}$ of a gene's assayed residues, the likelihood ratio of each call type with respect to the assay outcome is certified on a disjoint part $\mathcal{B}$, and a mixture check asks whether the unlabelled target scores are a re-weighting of the two classes of $\mathcal{A}$. On the score axis the two thresholds give three outcomes.

experiments (Table 1). Second, gene-level labelled sets are small and substitutions at one residue are dependent: when the population likelihood ratio was known, thresholds selected on percentile-bootstrap bounds respected every claimed bound in only 57.6% of resampled data sets (Section 4.4).

We introduce CertiVar (Figure 1), which turns a predictor score into loss-of-function (LoF), functional or no calls by two cut-offs. Per-class (Mondrian) conformal thresholds, set from SGE or DMS labels with one variant per residue, bound how often each class is mis-called. A certification step bounds the likelihood ratio of each call type with respect to the assay outcome, valid after a threshold has been selected from a grid. Neither step needs the LoF share of the variants being called, provided their within-class score distribution matches that of the assayed variants; a mixture-consistency check tests a necessary condition for this on the unlabelled targets. The certified quantity is how strongly a call predicts the assay result; it is evidence about pathogenicity only to the extent that the assay is clinically validated. Our contributions are: (1) certified likelihood ratios for the two call types of an abstaining classifier on residue-clustered data, valid after threshold selection with a Bonferroni correction or with fixed-sequence testing, per level or simultaneously over levels, together with a coverage guarantee after selection and disjoint call sets (Section 3); (2) a mixture-consistency check of class-conditional invariance that needs no target labels, kept its 5% level on random splits and rejected every tested set of unassayed VUS outside the assayed residues (Section 4.5); (3) a benchmark against known population likelihood ratios in which exact and Katz bounds held while percentile-bootstrap bounds and point estimates overstated evidence (Section 4.4); and (4) evidence on five genes and three predictors that marginal and prior-reweighted conformal prediction lose coverage of one class when the calibration LoF share moves from 5% to 50%, while per-class coverage holds as the theory predicts (Section 4.2).

2 Related work

ClinGen maps predictor scores to ACMG/AMP evidence strengths through local likelihood ratios estimated against ClinVar labels, with a one-sided bootstrap bound [2]; the approach has been extended to more predictors [6], to assay scores [5, 8, 9] and to gene- and domain-specific calibration [7, 10, 11], with uncertainty from the bootstrap [2, 6] or sign-test bounds on resampled likelihood ratios [7]. Split conformal prediction gives finite-sample coverage [12]; its class-conditional (Mondrian) form controls coverage within each class [12, 13], and clustered conformal prediction pools classes with similar scores when each class has few labels [14]; order-statistic thresholds give training-conditional control [15–17]; keeping one member per group handles clustered data [18]; and abstaining classifiers trade the share of inputs classified against error [19]. Choosing a threshold from candidates with exact binomial tail bounds and a union bound is the basis of selective classification with guaranteed risk [20] and of risk-controlling prediction sets [21]; fixed-sequence testing over an ordered grid is the Learn-then-Test template [22]. Harrizi et al. reported split-conformal

prediction sets for missense predictors on ClinVar benchmarks at 90% coverage [23]; CertiVar instead controls coverage within each class, calibrates on the gene's assay labels and certifies the likelihood ratio of the calls. Large-sample intervals for a ratio of binomial proportions [24, 25] are narrower than our ratio of exact endpoints but carry no finite-sample guarantee; Section 4.4 compares the Katz interval. Label-shift methods adjust posteriors or conformal weights to an estimated target prior [26–28]; class-conditional quantities need no prior, provided the target satisfies class-conditional invariance (Assumption 2), which Section 3 tests. Relative to these methods, CertiVar certifies a ratio of class-conditional call rates for both call types of an abstaining rule, on residue-clustered assay labels and on the OddsPath scale, tests the invariance it relies on, and measures against known population likelihood ratios how often point estimates and bootstrap bounds overstate evidence.

3 Method

Setting. Each missense variant has a predictor score s, oriented so that higher means more damaging, and an assay label $Y \in \{0, 1\}$, with $Y = 1$ for loss of function (LoF). A *residue* is a protein position of one gene; the substitutions at a residue share structural context and form one cluster. A gene's assayed missense variants are split by residue into a calibration part $\mathcal{A}$ and a certification part $\mathcal{B}$; the variants to be called (for example VUS) form the target. We write $q(s) \in [0, 1]$ for an increasing map of the score that puts the three predictors on one scale (here Platt scaling [29] fitted to confidently labelled ClinVar variants, which the fixed-cut baselines of Section 4.1 also use). Because q is strictly increasing, every CertiVar call is the same on q as on s, and no ClinVar label affects any call.

Per-class thresholds. The nonconformity of label 1 is $r_1 = 1 - q$ and of label 0 is $r_0 = q$. From each residue in $\mathcal{A}$ we keep one variant chosen uniformly at random. For class k, let n_k be the number of kept variants with label k and $r^{(k)}_{(1)} \le \cdots \le r^{(k)}_{(n_k)}$ their sorted nonconformities. The threshold is $\hat{t}_k = r^{(k)}_{(j_k)}$ (∞ if $j_k > n_k$), where j_k is the in-expectation rank or the probably approximately correct (PAC) rank at confidence level δ_{P} ($j_k = n_k + 1$ if no j qualifies):

$$j_k = \lceil (n_k + 1)(1 - \alpha_k) \rceil \text{ (in expectation)} \quad \text{or} \quad j_k = \min\{\, j : \; F_{\mathrm{Beta}(j,\, n_k - j + 1)}(1 - \alpha_k) \le \delta_{\mathrm{P}} \,\} \text{ (PAC).} \quad (1)$$

The prediction set is $C(q) = \{k : r_k \le \hat{t}_k\}$; CertiVar calls LoF if $C = \{1\}$, functional if $C = \{0\}$, and abstains otherwise. In effect it places two cut-offs on the score, $\hat{t}_0$ and $1 - \hat{t}_1$, each an order statistic of one class: a variant below both is called functional, one above both is called LoF, and one between them is not called (Figure 1). A functional call on a LoF variant requires $1 \notin C$, so per-class coverage bounds both error rates.

Certified likelihood ratios. The ACMG/AMP guidelines [3] grade evidence in qualitative categories; in their Bayesian adaptation, supporting, moderate, strong and very strong evidence towards pathogenicity correspond to a likelihood ratio above 2.1, 4.3, 18.7 and 350, and supporting, moderate and strong evidence towards benignity to one below 0.48, 0.23 and 0.053 [4, 5]. We reuse these values as a reporting scale, test levels up to strong, and write, for example, *LoF-supporting* for LoF calls whose likelihood ratio exceeds 2.1. Our likelihood ratios are defined with respect to the assay outcome Y and averaged over all variants that receive a call, as OddsPath is for a functional assay [5]; they are not the local, score-specific ratios that ClinGen uses for the computational criteria PP3/BP4 [2]. A level thus describes the call class as a whole, need not hold for each call, and must not be combined with the same assay's functional evidence (PS3/BS3) or with PP3/BP4 from the same predictor. For thresholds t, the call rates are $\lambda_k(t) = \Pr(\text{call LoF} \mid Y = k)$ and $\phi_k(t) = \Pr(\text{call functional} \mid Y = k)$, and $\mathrm{LR}^+ = \lambda_1/\lambda_0$, $\mathrm{LR}^- = \phi_1/\phi_0$.

For each α on a grid G we compute thresholds on $\mathcal{A}$ ($\alpha_0 = \alpha_1 = \alpha$), call one variant per residue of $\mathcal{B}$, and bound the four rates by one-sided Clopper–Pearson bounds [30] at level $\epsilon = \delta/(2|G|)$: for x calls among n variants of a class, the lower bound is $F^{-1}_{\mathrm{Beta}(x,\, n-x+1)}(\epsilon)$ (0 if $x = 0$) and the upper bound $F^{-1}_{\mathrm{Beta}(x+1,\, n-x)}(1-\epsilon)$ (1 if $x = n$, including $n = 0$). Writing $\underline{\lambda}_1, \overline{\lambda}_0, \overline{\phi}_1, \underline{\phi}_0$ for these bounds, the certified bounds are $L^+(\alpha) = \underline{\lambda}_1/\overline{\lambda}_0$ and $U^-(\alpha) = \overline{\phi}_1/\underline{\phi}_0$ ($U^- = \infty$, never meeting a level, if $\underline{\phi}_0 = 0$; $\overline{\lambda}_0 > 0$ always; for point estimates, $x/0 = \infty$ for $x > 0$, and $0/0$ is undefined and never selected). We write $\hat{t}(\alpha)$ for the pair $(\hat{t}_0, \hat{t}_1)$ at α and call the share

Algorithm 1 CertiVar for one gene and one evidence level (Bonferroni version)

Require: scores q; assay labels split by residue into $\mathcal{A}, \mathcal{B}$; grid G; δ; levels τ (LoF) and τ' (functional)

1: keep one variant per residue of $\mathcal{A}$ and of $\mathcal{B}$
2: **for** $\alpha \in G$ **do**
3: thresholds $\hat{t}_0, \hat{t}_1$ on the kept variants of $\mathcal{A}$ by (1); call the kept variants of $\mathcal{B}$
4: $L^+(\alpha) \leftarrow \underline{\lambda}_1/\overline{\lambda}_0$, $U^-(\alpha) \leftarrow \overline{\phi}_1/\underline{\phi}_0$ (Clopper–Pearson, level $\delta/(2|G|)$ each)
5: **end for**
6: $\hat{\alpha}^+ \leftarrow \arg\max_{\alpha \in G}\{\text{share called LoF on } \mathcal{B} : L^+(\alpha) > \tau\}$
7: $\hat{\alpha}^- \leftarrow \arg\max_{\alpha \in G}\{\text{share called functional on } \mathcal{B} : U^-(\alpha) < \tau',\ \alpha \geq \hat{\alpha}^+\}$ (over all $\alpha \in G$ if $\hat{\alpha}^+$ does not exist)
8: **output** LoF calls at $\hat{\alpha}^+$ and functional calls at $\hat{\alpha}^-$ with their certified levels (no calls of a type whose set is empty); abstain otherwise

of the kept variants of $\mathcal{B}$ that receive a call of a given type its *yield*. For each evidence level, Algorithm 1 returns the α with the largest yield among those whose certified bound meets the level. Because $\hat{t}_0$ and $\hat{t}_1$ both decrease weakly in α, no variant can be called LoF at one α and functional at another (Section 3.1). Line 7 requires, for functional calls, α at least as large as the LoF choice of the same strength, so that the two call types of one strength come from an ordered pair of thresholds; this is a reporting convention, not needed for validity, and can only lower functional yields. The *fixed-sequence* variant replaces the Bonferroni level $\delta/(2|G|)$ by $\delta/(2L)$, where L is the number of evidence levels tested for a call type, tests the grid in an order fixed in advance, and stops at the first α that does not meet the level (Theorem 2). We test LoF levels in increasing and functional levels in decreasing α, because few variants are called functional at small α. All thresholds and bounds cost $O(|G|\, n \log n)$ for n residues, about 5 ms per gene on a laptop CPU.

Checking class-conditional invariance. The guarantees transfer to a target set whose LoF share differs from the calibration data only if, within each class, target scores follow the calibration law (Assumption 2 below). This implies that the target score law is a mixture $\pi F_1 + (1-\pi)F_0$ of the two class laws for some $\pi \in [0,1]$. We test this necessary condition on one target variant per residue with $D = \min_\pi \sup_x \left|\hat{F}_T(x) - \pi\hat{F}_1(x) - (1-\pi)\hat{F}_0(x)\right|$, where $\hat{F}_T$, $\hat{F}_1$ and $\hat{F}_0$ are the empirical distribution functions of the target scores and of the two classes in $\mathcal{A}$, and π runs over a grid of step 0.005. The null distribution of D is approximated by a bootstrap that redraws both calibration classes and a target sample of the same size from the fitted mixture (300 resamples). A rejection shows that the target is not a re-weighting of the calibration classes, so the guarantees need not transfer, and we do not report calls for such a target set; acceptance does not establish Assumption 2.

3.1 Guarantees

Let Q be the law of a variant obtained by drawing a residue from the gene's population of assayed residues and then one of its variants uniformly at random (binary-labelled, non-splice variants outside the ClinVar labelled set). Every rate and likelihood ratio below is taken under Q, so it weights residues equally.

Assumption 1 (Clustered sampling)**.** Residues in $\mathcal{A}$, in $\mathcal{B}$ and of a test variant are independent draws from one residue population; within a residue, the kept or tested variant is chosen uniformly at random; q does not depend on these data.

Assumption 2 (Class-conditional invariance)**.** Target variants have, for $k \in \{0,1\}$, the same law of q given $Y = k$ as a test variant under Assumption 1; their LoF share $\pi = \Pr(Y = 1)$ may differ.

Proposition 1 (Per-class coverage)**.** *Under Assumption 1, for a fixed α_k, with the in-expectation rank and conditionally on n_k, $1 - \alpha_k \leq \Pr(k \in C \mid Y = k, n_k) \leq 1 - \alpha_k + 1/(n_k + 1)$, the upper bound holding when there are no ties.*

Proposition 2 (High-probability per-class coverage)**.** *Under Assumption 1, for a fixed α_k and with the PAC rank, $\Pr(k \in C \mid Y = k, \hat{t}_k) \geq 1 - \alpha_k$ with probability at least $1 - \delta_{\mathrm{P}}$ over the calibration data for each k, and for both classes with probability at least $1 - 2\delta_{\mathrm{P}}$.*

Proposition 3 (Coverage after selection)**.** *Under Assumption 1, for any $\hat{\alpha} \in G$ chosen from data independent of the test variant (for example by Algorithm 1), the in-expectation rank gives $\Pr(k \in C \mid Y = k) \geq 1 - \max G$; the PAC rank at level $\delta_{\mathrm{P}}/|G|$ gives $\Pr(k \in C \mid Y = k, \hat{t}_k) \geq 1 - \hat{\alpha}$ for each k with probability at least $1 - \delta_{\mathrm{P}}$.*

Theorem 1 (Certified likelihood ratios)**.** *Under Assumption 1, conditionally on $\mathcal{A}$ and with probability at least $1 - \delta$ over $\mathcal{B}$, $\mathrm{LR}^+(\hat{t}(\alpha)) \geq L^+(\alpha)$ for every $\alpha \in G$ simultaneously; the same holds for $\mathrm{LR}^- \leq U^-$ with a second δ. Because the event does not depend on the level, each call type returned by Algorithm 1 reaches its certified level at every evidence level reported, with probability at least $1 - \delta$ (both call types with probability at least $1 - 2\delta$), whichever α is selected.*

Theorem 2 (Fixed-sequence certification)**.** *Under Assumption 1, fix a level τ, a confidence δ' and an order of G chosen in advance. Reject $H_\alpha : \mathrm{LR}^+(\hat{t}(\alpha)) \leq \tau$ when $\underline{\lambda}_1/\overline{\lambda}_0 > \tau$ with both endpoints at level $\delta'/2$, stop at the first α not rejected, and select the rejected α with the largest yield; the rejected set is a prefix of the order. (a) Conditionally on $\mathcal{A}$, the selected LoF calls have $\mathrm{LR}^+ > \tau$ with probability at least $1 - \delta'$. (b) If each of L levels is tested in this way with $\delta' = \delta/L$, every level claimed for LoF calls holds simultaneously with probability at least $1 - \delta$. The same holds for functional calls with $\overline{\phi}_1/\underline{\phi}_0 < \tau'$, and for both call types together with probability at least $1 - 2\delta$.*

Corollary 1 (No prior needed)**.** *Under Assumptions 1 and 2, for target variants drawn residue first as under Q, Propositions 1–3 and Theorems 1–2 hold whatever their LoF share $\pi \in (0, 1)$: for a fixed α, LoF variants are called functional at rate at most α_1, and the certified likelihood ratios with respect to the assay outcome need no prior. For any $\alpha^+, \alpha^- \in G$, the LoF calls at α^+ and the functional calls at α^- are disjoint.*

Propositions 1 and 2 are the class-conditional and training-conditional conformal guarantees [12, 13, 15, 16], here under the residue-first law Q. Theorem 1 uses the union-bound template of risk control [20, 21] and Theorem 2 the fixed-sequence argument of Learn-then-Test [22]; both are applied here to a *ratio* of two class-conditional rates of an abstaining rule, whose four counts are binomial given $\mathcal{A}$ and the labels in $\mathcal{B}$. What makes selection safe for coverage and for the two call types together is monotonicity: both ranks in (1) are non-increasing in α, so coverage at a selected α is at least coverage at $\max G$, and a variant cannot be called LoF at one grid point and functional at another. Proofs are in Appendix S1.

Scope. After selection, per-class coverage is guaranteed only at $1 - \max G$ with the in-expectation rank; we report the coverage achieved in Section 4.4. Theorem 2 certifies levels, not numerical bounds. Assumption 1 treats assayed residues as independent draws; our random splits draw them without replacement from a finite set, which the oracle design of Section 4.4 avoids, and a contiguous region need not be exchangeable with the rest; Assumption 2 holds only approximately for VUS, on which AlphaMissense discriminates less well (Section 4.1), and not for variants whose assay result would be intermediate.

4 Experiments

4.1 Data and protocol

We took every GRCh38 missense single-nucleotide variant in BRCA1, BRCA2, BAP1, RAD51C and MSH2 that is in ClinVar [1] (release of 26 October 2026) or was assayed, annotated each on its MANE Select transcript with the Ensembl Variant Effect Predictor [36] (AlphaMissense [37], REVEL [38]) and computed ESM-2 650M [39] log-likelihood ratios by wild-type marginals [40]. Labels come from five MaveDB [41] score sets (Table 1; accessions in Appendix S5): the authors' classes for BRCA1, BAP1 and RAD51C and the authors' threshold for MSH2 (positive loss-of-function score [35]). The BRCA2 score set has no deposited classes, so we fitted Gaussians to the assay's synonymous and nonsense variants (LoF if the posterior is at least 0.95, functional if at most 0.05). Applied to the genes with the authors' classes, this control rule agreed on all 1,762 BRCA1 variants confident under both, disagreed on 216 of 4,023 for BAP1 (5.4%), and left 76.8% of the authors' functional RAD51C variants intermediate; Section 4.6 repeats the main experiments under these alternative label rules. We excluded the 729 assayed variants in splice regions and set aside the 1,274 intermediate variants. One Platt map per predictor, pooled over genes, used the 859 ClinVar missense variants with at least two review stars and a pathogenic or likely pathogenic (P/LP) or benign or likely

Table 1: Datasets. *ClinVar labelled*: missense single-nucleotide variants (SNVs) with ≥2 review stars and a pathogenic or likely pathogenic (P/LP) or benign or likely benign (B/LB) call, used only to put the scores on a common scale, and the number that are P/LP. *Assayed*: missense SNVs with a binary assay label outside that set, their residues and their LoF share; *assayed VUS*: the subset classified VUS in ClinVar. Shares are variant-level. Labels: the authors' classes (authors) or a rule fitted to the assay's synonymous and nonsense controls (controls; Section 4.1).

			ClinVar labelled		Assayed			Assayed VUS	
Gene	Assay	Labels	n	P/LP	n	Residues	LoF	n	LoF
BRCA1	SGE [31]	authors	345	156	1,617	307	0.161	272	0.173
BRCA2	SGE [32]	controls	309	94	3,435	676	0.130	704	0.207
BAP1	SGE [33]	authors	14	6	4,294	696	0.213	1,069	0.166
RAD51C	SGE [34]	authors	13	10	2,225	355	0.259	550	0.396
MSH2	DMS [35]	authors	178	121	2,906	893	0.060	1,542	0.067
Total			859	387	14,477	2,927	0.164	4,137	0.167

Table 2: Per-class coverage at $\alpha = 0.1$ (residue-weighted; 50 random halvings of each gene's residues). Worst class: the smaller of the mean LoF and mean functional coverage of a gene–predictor pair. Last column: worst pair when the calibration half is subsampled to a LoF share of 0.05 and tested at the natural share (100 halvings); fixed cut-offs use no calibration data.

		Worst-class coverage				Worst pair,
Method	Pairs	mean	worst pair	pairs < 0.85	Abstention	share 0.05
Per-class (ours)	15	0.901	0.896	0	0.35	0.899
Pooled conformal	15	0.836	0.545	5	0.42	0.113
Label-shift weighted, true share	15	0.836	0.559	5	0.42	0.598
Label-shift weighted, EM share	15	0.722	0.039	9	0.36	0.038
Fixed tiers ($q \geq 0.9$, $q \leq 0.1$)	15	0.871	0.663	5	0.53	–
AlphaMissense classes	5	0.725	0.540	4	0.11	–
REVEL PP3/BP4 thresholds	5	0.700	0.445	5	0.42	–

benign (B/LB) call (45.1% P/LP), which are excluded from every evaluation set, leaving 14,477 assayed variants at 2,927 residues. Pooled over genes, AlphaMissense has an area under the ROC curve (AUROC) of 0.946 (95% residue-bootstrap interval 0.922–0.965) on the ClinVar set, 0.840 (0.829–0.852) on the other assayed variants and 0.779 (0.757–0.799) on assayed VUS. Baselines for the calls are pooled (marginal) split conformal prediction; label-shift-weighted conformal prediction [28] with the true target share or its expectation–maximization (EM) estimate [26]; fixed tiers ($q \geq 0.9$ LoF, $q \leq 0.1$ functional); the published AlphaMissense classes (> 0.564, < 0.34); and the REVEL PP3/BP4 thresholds (≥ 0.644, ≤ 0.290) [2]. None claims per-class coverage, and the published cut-offs target pathogenicity, so we judge them only as assay classifiers. Coverage is residue-weighted as in Q; *worst-class coverage* is the smaller of the mean LoF and mean functional coverages over repetitions (Figure 2B uses the smaller of the two in each split, and Table 4 in each target set). Every split is by residue (designs in Table S7).

4.2 Per-class coverage and a shift in the calibration LoF share

Over 50 random halvings of each gene's residues (calibrate on one half, test on the other), the in-expectation rule at $\alpha = 0.1$ gave worst-class coverage between 0.896 and 0.908 in all 15 gene–predictor pairs, none more than one Monte Carlo standard error (0.002–0.010) below the target (Table 2, Figure 2A; per pair in Table S8). Pooled conformal fell to 0.545 on LoF variants and below 0.85 in 5 pairs, because a marginal threshold controls coverage averaged over the two classes, not within each (in MSH2, with 6% LoF variants, it was the functional class that fell marginally below 0.90); the published cut-offs covered functional variants as little as 0.540 (AlphaMissense, BAP1) and 0.445 (REVEL, MSH2). The per-class rule abstained on 35%

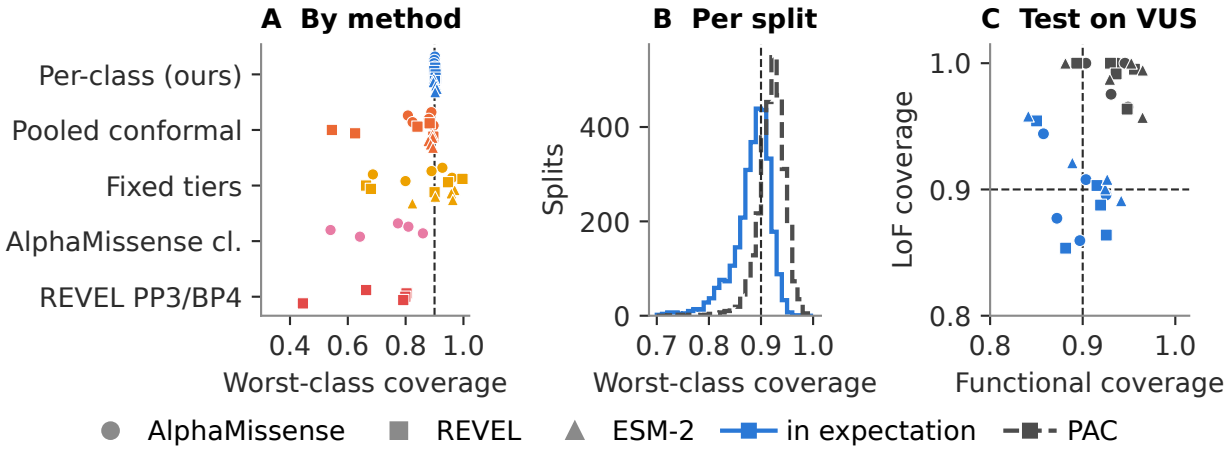


Figure 2: Per-class coverage at $\alpha = 0.1$. Marker shape: predictor (panels A, C); colour and line style: threshold rule (panels B, C). (A) Worst-class coverage for each gene–predictor pair (50 splits); AlphaMissense cl.: the published AlphaMissense classes. (B) Smaller of the two per-class coverages in each test half, pooled over pairs (3,000 splits per rule). (C) Coverage on assayed VUS with thresholds set on non-VUS variants at other residues (50 repetitions).

of variants on average (17–47%), against 42% for pooled conformal and 11% for the AlphaMissense classes. Calibrating on all variants of the calibration half instead of one per residue used about five times as many labels and abstained less (30.4% against 35.5%); but in a separate run of 50 halvings that compared the two, worst-class coverage fell below 0.90 in 12 of 15 pairs (lowest 0.885), against 5 (lowest 0.897) for one variant per residue. All-variant calibration weights variants while the test weights residues; weighting each residue equally left 1 pair below 0.90 (lowest 0.898) but abstained more (44.6%), which points to the weighting rather than the dependence, so we keep one variant per residue.

To compare the methods under a prevalence shift, we subsampled each calibration half to a LoF share of 0.05 to 0.5 and tested at the natural share (100 repetitions; Figure 3A). Subsampling by class leaves each class's score law unchanged, so Corollary 1 predicts the per-class result; the question is what happens to the baselines. The per-class rule kept worst-class coverage between 0.897 and 0.925 in every pair at every share (0.891–0.925 in a separate run balanced at 50% LoF), with abstention rising from 37% to 83% as LoF variants became scarce. Pooled conformal calibrated at share 0.05 covered LoF variants at 0.113 in its worst pair, and label-shift-weighted conformal fell to 0.53–0.60 with the true target share and to 0.04 with the EM estimate, which missed the LoF share by 0.255 on average because ClinVar pathogenicity and assay loss of function differ in definition as well as in class share.

4.3 Reliability across splits and a real change of population

With 17 to 93 LoF residues per calibration half, a single split can miss the in-expectation guarantee: over 200 splits per pair, LoF coverage on the test half was below 0.90 in 36.4% of splits and functional coverage in 43.6% (Figure 2B). PAC thresholds ($\delta_P = 0.1$) reduced these to 7.9% and 10.8% at the price of abstaining on 58.4% of variants against 35.4%. A finite test half adds sampling noise to the coverage that Propositions 1 and 2 bound. Computed exactly on all of a gene's residues for calibration samples drawn from them (200 draws per pair), PAC coverage fell below 0.90 in 4.8% (LoF) and 7.1% (functional) of draws and in at most 10.0% in any pair, within δ_P; in-expectation coverage did so in 34.3% and 42.9%. Calibrated on non-VUS variants of one half of each gene's residues and applied to the assayed VUS of the other half (residue-weighted LoF share 17.0% in the calibration sample and 19.8% among the VUS), in-expectation thresholds kept LoF coverage between 0.854 and 0.958 and functional coverage between 0.841 and 0.942, each below 0.90 in 7 of 15 pairs (Figure 2C); PAC thresholds kept LoF coverage at or above 0.957, but functional coverage fell below 0.90 in 2 of 15 pairs (lowest 0.881), and they abstained on 81.7% of VUS; on VUS, Assumption 2 holds only approximately.

Table 3: Threshold selection judged against the population likelihood ratio (oracle design, grid G_4, $\delta = 0.05$, 1,500 draws; all rules on the same draws). Guarantee: finite-sample validity after selection, for every level of a call type at once or for each level separately. Bounds held: draws in which every claimed bound was respected (fixed-sequence rules certify a level, not a bound; a point estimate is not a bound). Levels held: draws in which every claimed level held. Last four columns: share of draws certifying each level (%); no rule with a guarantee certified a strong level.

	Finite-sample	Bounds	Levels	LoF calls		Functional calls	
Selection rule	guarantee	held	held	supp.	mod.	supp.	mod.
Certified (ours), Bonferroni	yes, all levels	99.7	99.9	86.7	13.3	44.9	1.5
Certified (ours), fixed sequence	yes, all levels	–	99.9	74.9	14.4	43.8	1.5
Fixed sequence, each level at δ	yes, per level	–	99.7	87.2	30.5	61.9	3.9
Clopper–Pearson, uncorrected	no	97.9	99.5	95.9	35.1	68.0	6.9
Katz interval, Bonferroni	no	98.0	99.7	99.3	52.5	52.4	0.2
Percentile bootstrap	no	57.6	72.5	99.9	78.1	90.0	45.1
Percentile bootstrap, Bonferroni	no	65.3	75.1	98.1	52.3	74.3	36.8
Percentile bootstrap, +0.5 counts	no	75.4	88.2	99.9	76.6	89.3	37.3
Point estimate	no	15.5	42.1	100.0	98.2	99.8	75.7

4.4 Certified likelihood ratios

Treating each gene's assayed residues as the population, we drew calibration (30%) and certification (50%) residues with replacement, 100 times per pair, ran Algorithm 1 with $\delta = 0.05$ on the four-point grid $G_4 = \{0.05, 0.10, 0.20, 0.30\}$, and computed the population likelihood ratio of every selection exactly. All rules were applied to the same draws (Table 3). Because Assumption 1 holds here by construction, this design checks the certificates' arithmetic and the competitors' failures; the held-out regions of Section 4.6 test what happens when it does not. Every bound claimed in a draw was respected in 99.7% of the 1,500 draws for the certified rule (Theorem 1 guarantees at least 90%). Percentile-bootstrap bounds were all respected in only 57.6% of draws, 65.3% with a Bonferroni correction and 75.4% with 0.5 added to every count; 59% of bootstrap failures came from certification samples with no false call of that type, where the bound failed 89% of the time. Point estimates claimed levels that all held in only 42.1% of draws; their strong LoF claims held in 50 of 235 (Figure S3). Two rules without a guarantee after selection, uncorrected Clopper–Pearson bounds and Katz intervals with a Bonferroni correction, held in 97.9% and 98.0% of draws and certified more often (Clopper–Pearson at every level, Katz at three of the four): Katz bounds reached LoF-moderate in 52.5% of draws against 13.3%. The exact certificate therefore costs yield, which buys a guarantee that holds at every sample size.

CertiVar certified LoF-supporting calls in 86.7% of draws (calling 25.2% of variants), functional-supporting calls in 44.9% (61.4% called), LoF-moderate in 13.3% and functional-moderate in 1.5%, and never a strong level. Per-class coverage at the selected α averaged 0.81 (LoF) and 0.80 (functional), above the guaranteed 0.70 (Proposition 3): selecting for evidence strength trades coverage for confidence. Claimed levels held in 99.9% of certified claims when the likelihood ratio was weighted by variant rather than by residue. Fixed-sequence testing with the level-simultaneous guarantee of Theorem 2(b) traded supporting for moderate evidence: on G_4 it certified LoF-moderate in 14.4% of draws against 13.3% but LoF-supporting in 74.9% against 86.7%, and on a 20-point grid (G_{20}, 20 equally spaced values from 0.02 to 0.40) LoF-moderate in 12.7% against 6.4% but LoF-supporting in 50.3% against 74.8% (Figure 3B). Testing each level at the full δ, which guarantees one level at a time, reached LoF-moderate in 30.5% of draws on G_4; the testing order also matters, since functional levels tested in increasing α reached functional-supporting in 24.7% of draws against 61.9%. On held-out fifths of the residues, empirical likelihood ratios respected the certified bounds in 98.7% of 2,091 cases (bootstrap: 76.8%); on VUS at other residues, LoF-supporting held in all 443 certified claims and LoF-moderate in 25 of 28. Against ClinVar labels (exploratory, not independent of the assays), LoF calls had likelihood-ratio lower bounds of 4.9–19.6 (BRCA1), 7.6–9.4 (BRCA2) and 2.6–3.2 (MSH2) (Table S4).

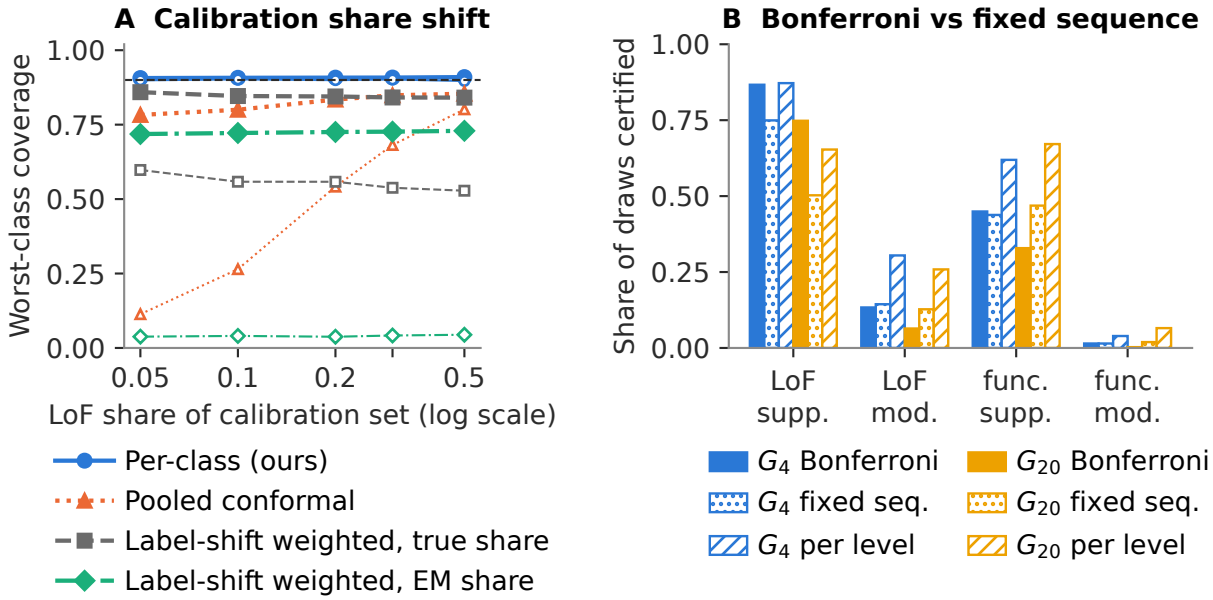


Figure 3: (A) Worst-class coverage when the calibration half is subsampled to a given LoF share (0.05, 0.1, 0.2, 0.3 and 0.5, log scale): mean over 15 pairs (thick) and worst pair (thin, open markers). (B) Share of oracle draws certifying each level with a Bonferroni correction, with fixed-sequence testing simultaneous over levels (fixed seq.), or with fixed-sequence testing of each level at the full δ (per level), on the four-point (G_4) and 20-point (G_{20}) grids.

Table 4: Mixture-consistency check of class-conditional invariance (Section 4.5; one target variant per residue, 300 bootstrap resamples, rejection at $p < 0.05$). Coverage: mean over target sets of the smaller per-class coverage of in-expectation thresholds ($\alpha = 0.1$) set on the same calibration data, where target labels exist.

		Target	Rejected	Worst-class coverage	
Target set	Assumption 2	sets	(%)	accepted	rejected
Random half of the residues	holds by design	1,500	4.7	0.880	0.880
Assayed VUS at other residues	approximately	750	8.5	0.845	0.816
Held-out contiguous quarter	not assumed	1,500	43.9	0.837	0.788
Another gene's calibration data	not assumed	300	77.3	0.745	0.644
Unassayed VUS, other residues	unknown (no labels)	120	100.0	–	–

4.5 Checking invariance, and where CertiVar applies

The mixture-consistency check of Section 3 rejected 4.7% of random halves (3.0–7.7% by gene), where Assumption 2 holds by design, so it keeps close to its 5% level (Table 4). It rejected 43.9% of held-out contiguous quarters, 77.3% of transfers from another gene's calibration data and every set of unassayed ClinVar VUS at residues the BRCA1 and BRCA2 assays did not reach; on those VUS the thresholds abstained on only 8.9% of variants (mean over predictors and the two genes), against 35% on assayed residues. Where labels exist, rejected targets had lower coverage than accepted ones (0.788 against 0.837 for quarters, 0.644 against 0.745 across genes). The check has little power for assayed VUS (8.5% rejected), whose accepted sets had a mean worst-class coverage of 0.845 against 0.880 for random halves on the same measure, so passing it is not evidence that the guarantees transfer. Across the five genes, 147 ClinVar missense VUS were unassayed but lay at assayed residues, 105 of them in MSH2 (Tables S5 and S6). Even these need not be exchangeable with the assayed variants, since they are the ones the assay missed: for MSH2 the check rejected 40% of samples with AlphaMissense and 35% with REVEL, but none with ESM-2, whose thresholds gave certified LoF-supporting calls to 19.1 and functional-supporting calls to 5.0 of the 105 on average over 20 splits; for the other two predictors, Table S6 averages only over the splits the check accepted. Of the 4,044 unassayed

VUS at other residues, the 4,010 in BRCA1 and BRCA2 are not called, because the check rejected every sample of them; the other 34 are too few to check.

4.6 How much data, from where, and with which labels

Calibration data had to come from the same gene: calibrating on the other four genes left 8 of 15 pairs below 0.85 (worst 0.487), consistent with gene-specific evidence strength [10, 11]. Calibrating on three contiguous quarters of a protein and testing on the fourth gave worst-class coverage of 0.860–0.923, and 88.8% of 1,891 certified bounds held against the empirical likelihood ratios of the held-out quarter, below the 95% that Theorem 1 guarantees under Assumption 1 (those ratios carry their own sampling noise). With 150 calibration residues drawn at random, coverage averaged over genes met the target with 40.5% abstention (worst pair 0.890; 82.8% abstention with 25 residues). Coverage was lower on the third of variants nearest the assay class boundary (0.865 LoF, 0.873 functional). Intermediate variants, which no guarantee covers, were called LoF in 19.6% and functional in 36.6% of cases. Relabelling BRCA2, BAP1 and RAD51C with alternative rules (Table S3) changed yields but not validity: the lowest per-class coverage over predictors stayed between 0.894 and 0.902, and certified bounds held in 2,530 of 2,533 oracle selections.

5 Discussion

Thresholds selected on point estimates or percentile-bootstrap bounds often claimed more evidence than held. Because these are call-averaged ratios, not the local ratios of ClinGen calibrations, this does not show that published PP3/BP4 thresholds are wrong, only that point estimates and bootstrap bounds can overstate evidence at single-gene sample sizes. Bounds built as confidence intervals for the ratio, exact or Katz, held in at least 97.9% of draws; the exact certificate is the conservative choice that keeps a guarantee at every sample size. CertiVar makes such claims checkable without an estimate of the share of damaging variants, and its mixture check says when a target set lies outside what the assay data can certify. It applies to genes with a functional assay: to audit whether a predictor's claimed evidence strength holds against the assay, and to call the variants the assay missed at assayed residues (147 ClinVar VUS here), for predictors whose scores pass the check. For the 4,010 unassayed BRCA1 and BRCA2 VUS at other residues, and for AlphaMissense and REVEL in part of the MSH2 samples, the check declined. The cost is conservatism: certified evidence was mostly supporting-level, and PAC thresholds abstained on 58% to 82% of variants.

Limitations. The labels are assay classes, not clinical pathogenicity [5], and the ACMG/AMP levels are reused here as a reporting scale; clinical use requires an assay validated against clinical classifications, as its authors did for BRCA1 [31] and, with their own classes, for BRCA2 [32]. The BRCA2 score set has no deposited classes, so our BRCA2 labels come from the assay's controls (sensitivity in Table S3). A certified level describes the calls as a class and weights residues equally, so a borderline call can carry less evidence than its class. The guarantees assume residues exchangeable with the calibrated ones; they do not cover intermediate, splice-region or out-of-region variants. Intermediate variants are a hazard, because whether a VUS would score intermediate is unknown before it is assayed, and 19.6% of them were called LoF. The mixture check tests only a necessary condition and missed most VUS shifts. Results cover five cancer-predisposition genes, four of them in DNA repair. None of the predictors was trained on these assay labels, but REVEL was trained on labelled variants and the AlphaMissense classes were set on ClinVar, so both may overlap our evaluation sets [38, 42]; ESM-2, trained without variant labels, gave the same coverage and validity.

6 Conclusion

CertiVar turns a variant-effect predictor and a few hundred assayed residues of a gene into LoF, functional or no calls with finite-sample guarantees on per-class error rates and on call-averaged likelihood ratios for the assay after threshold selection, without an estimate of how common damaging variants are, and checks whether a target set can inherit them. Code, data and results (release v1.1.0): `https://github.com/aaryavvatts-lab/certivar`.

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Appendix

Section, equation, figure and table numbers without the prefix S refer to the main text. Contents: S1 proofs of Propositions 1–3, Theorems 1–2 and Corollary 1; S2 simulation (Fig. S1) and why marginal thresholds do not control the minority class; S3 calibration size, region and gene transfer (Fig. S2A–C), per-gene coverage (Table S1) and prior estimation (Table S2); S4 strata, assay-label sensitivity (Table S3), calls judged against ClinVar labels (Table S4), ClinVar VUS by assay status and the calls on them (Tables S5–S6) and the oracle comparison (Fig. S3); S5 implementation and the comparison rules; S6 experimental designs (Table S7) and all baselines by gene and predictor (Table S8).

S1 Proofs

Assumptions, propositions, theorems and the corollary are stated in Section 3.1; Q, C, $\hat{t}_k$, the call rates λ_k, ϕ_k and their bounds, L^+ and U^- are defined in Section 3.

Throughout, fix a class k and write F_k for the distribution function of the nonconformity r_k of a variant drawn from $Q(\cdot \mid Y = k)$. By Assumption 1, the kept calibration variants (one per residue of $\mathcal{A}$) and the variant to be called are independent draws from Q; conditionally on all labels, the n_k kept variants of class k and a class-k test variant are independent draws from $Q(\cdot \mid Y = k)$.

Proposition 1. Conditionally on the labels, the $n_k + 1$ nonconformities of the class-k kept variants and of the test variant are exchangeable, so with ties broken at random the rank of the test value is uniform on $\{1, \ldots, n_k + 1\}$ [12, 43]. The test value is at most the j_k-th smallest calibration value with probability $j_k/(n_k + 1)$, which lies in $[1 - \alpha_k,\ 1 - \alpha_k + 1/(n_k + 1))$ for $j_k = \lceil (n_k + 1)(1 - \alpha_k) \rceil$; ties only increase the probability, which gives the lower bound in general. If $j_k > n_k$ then $\hat{t}_k = \infty$. Since $k \in C$ exactly when $r_k \le \hat{t}_k$, the claim follows. □

Proposition 2. Conditionally on the labels, the coverage of the threshold is $F_k(\hat{t}_k) = F_k(r^{(k)}_{(j)})$. If F_k is continuous, $F_k(r^{(k)}_{(j)})$ is the j-th order statistic of n_k independent uniforms and has the $\mathrm{Beta}(j, n_k - j + 1)$ distribution, so $\Pr\{F_k(\hat{t}_k) < 1 - \alpha_k\} = F_{\mathrm{Beta}(j, n_k - j + 1)}(1 - \alpha_k) \le \delta_{\mathrm{P}}$ by the choice of j_k [15]. If F_k has atoms, write $r = F_k^{-1}(U)$ with U uniform and F_k^{-1} the generalized inverse; then $r^{(k)}_{(j)} = F_k^{-1}(U_{(j)})$ and $F_k(F_k^{-1}(u)) \ge u$, so $F_k(r^{(k)}_{(j)}) \ge U_{(j)}$ and the bound still holds. If no $j \le n_k$ qualifies, $\hat{t}_k = \infty$ and coverage is 1. □

Proposition 3. Fix $\mathcal{A}$. Both ranks in (1) are non-increasing in α (the in-expectation rank because $\lceil (n_k + 1)(1 - \alpha) \rceil$ is, the PAC rank because $F_{\mathrm{Beta}(j, n_k - j + 1)}(1 - \alpha)$ decreases in α for each j, so the smallest qualifying j cannot increase), so $\hat{t}_k(\alpha)$ is non-increasing in α and the set $\{r_k \le \hat{t}_k(\alpha)\}$ shrinks as α grows. Hence, for any $\hat{\alpha} \in G$ that depends on $\mathcal{A}$ and $\mathcal{B}$ but not on the test variant, $\Pr(k \in C_{\hat{\alpha}} \mid Y = k, \mathcal{A}, \mathcal{B}) \ge \Pr(k \in C_{\max G} \mid Y = k, \mathcal{A})$, and taking expectations over $\mathcal{A}$ and Proposition 1 at $\alpha_k = \max G$ give the first claim. For the PAC rank at level $\delta_{\mathrm{P}}/|G|$, Proposition 2 applied to each $\alpha \in G$ and a union bound give, with probability at least $1 - \delta_{\mathrm{P}}$, coverage at least $1 - \alpha$ for every $\alpha \in G$ simultaneously, in particular for $\hat{\alpha}$. □

Theorem 1. The thresholds $\hat{t}(\alpha)$ depend only on $\mathcal{A}$ and are fixed given $\mathcal{A}$. Conditionally on $\mathcal{A}$ and on the labels of the kept variants of $\mathcal{B}$, the class-1 kept variants of $\mathcal{B}$ are n_1 independent draws from $Q(\cdot \mid Y = 1)$, so the number of them called LoF is $\mathrm{Binomial}(n_1, \lambda_1(\alpha))$ with $\lambda_1(\alpha) = \Pr(\text{call LoF} \mid Y = 1)$, and likewise for the other three counts. Each one-sided Clopper–Pearson bound [30] at level $\delta/(2|G|)$ fails with probability at most $\delta/(2|G|)$; the conventions for $x = 0$, $x = n$ and $n = 0$ in Section 3 give the trivial bounds 0 and 1, which never fail. For LR^+ there are $2|G|$ such bounds ($\underline{\lambda}_1$ and $\overline{\lambda}_0$ for each α), so by the union bound all of them hold simultaneously with probability at least $1 - \delta$. On that event $\lambda_1(\alpha) \ge \underline{\lambda}_1(\alpha)$ and $\lambda_0(\alpha) \le \overline{\lambda}_0(\alpha)$ for every $\alpha \in G$, hence $\mathrm{LR}^+(\hat{t}(\alpha)) \ge L^+(\alpha)$ for every α (with $\mathrm{LR}^+ = \infty$ when $\lambda_0(\alpha) = 0$), including any $\hat{\alpha}^+$ chosen from these data. Since this probability bound holds for every $\mathcal{A}$, it also holds unconditionally. The argument for LR^- is identical with $\overline{\phi}_1$ and $\underline{\phi}_0$. The event does not involve the levels τ, so it covers every evidence level reported. □

Theorem 2. Fix $\mathcal{A}$, so the thresholds and the hypotheses $H_\alpha : \mathrm{LR}^+(\hat{t}(\alpha)) \le \tau$ are fixed. (a) If H_α is true and both endpoints $\underline{\lambda}_1(\alpha)$ and $\overline{\lambda}_0(\alpha)$ at level $\delta'/2$ hold, then $\underline{\lambda}_1 \le \lambda_1 \le \tau\lambda_0 \le \tau\overline{\lambda}_0$, so H_α is not rejected; hence a true H_α is rejected with probability at most δ'. Testing stops at the first non-rejection, so the rejected set is a prefix of the order. Let α^* be the first α in the order for which H_α is true (if there is none, every claim is true). The procedure claims the level at an α where H_α is true only if it rejects every hypothesis up to that α, which includes H_{α^*}; this has probability at most δ'. The selected α is one of the rejected ones, so the claimed level holds with probability at least $1 - \delta'$. (b) With $\delta' = \delta/L$ for each of L levels, a union bound over the levels shows that some claimed level fails with probability at most δ. The functional case is identical with $\overline{\phi}_1/\underline{\phi}_0 < \tau'$. □

Corollary 1. Every quantity bounded above is a probability conditional on the true class: $\Pr(k \in C \mid Y = k)$, the four call rates $\lambda_1, \lambda_0, \phi_1, \phi_0$, and their ratios. Under Assumption 2 these are the same for target variants as for a test variant under Assumption 1, whatever π is, so Propositions 1–3 and Theorems 1–2 transfer. The two call sets are disjoint for any $\alpha^+, \alpha^- \in G$: a LoF call at α^+ requires $q > \hat{t}_0(\alpha^+)$ and $1 - q \le \hat{t}_1(\alpha^+)$, a functional call at α^- requires $q \le \hat{t}_0(\alpha^-)$ and $1 - q > \hat{t}_1(\alpha^-)$, and both thresholds are non-increasing in α, so the first conditions contradict each other if $\alpha^- \ge \alpha^+$ and the second if $\alpha^- < \alpha^+$. A functional call on a LoF variant requires $1 \notin C$, so its rate is at most $1 - \Pr(1 \in C \mid Y = 1) \le \alpha_1$. None of these quantities involves π, so no prior is needed to certify them; this concerns the likelihood ratio with respect to the assay outcome, not evidence about pathogenicity. □

S2 Simulation

Scores were drawn per residue as $s = 2y + u_c + e$, with a residue effect $u_c \sim N(0, 0.5^2)$ shared by the five variants of a residue, unit Gaussian noise, and the class shared within a residue. A Platt calibrator was fitted on 600 variants at LoF share 0.45, conformal thresholds were set on another 600 (one variant per residue), and 3,000 test variants were drawn at LoF shares from 0.05 to 0.60, with 300 replicates per share (Figure S1). Per-class thresholds kept LoF coverage at the target at every share; pooled thresholds did not, and reweighting them [28] to the EM estimate of the share [26] kept marginal coverage near the target while LoF coverage fell further as LoF variants became rarer. Adjusting posteriors to the EM estimate removed the calibration error caused by the shift in this simulation, where the shift is a pure change of prior, unlike in the real data (Section 4.2).

Why marginal thresholds do not control the minority class. A pooled split-conformal threshold $\hat{t}$ is a quantile of the mixture $\pi_c F_1 + (1 - \pi_c)F_0$ of the two class-conditional nonconformity laws, weighted by the calibration share π_c. Its coverage of class 1 is $F_1(\hat{t})$, which is not controlled; when LoF variants are rare and harder to separate, $F_1(\hat{t}) < 1 - \alpha$ while marginal coverage is still near $1 - \alpha$. Reweighting to a known test prior [28] restores marginal coverage at that prior but does not control $F_1(\hat{t})$ either.

S3 Calibration size, region and gene transfer, and per-gene results

Figure S2 shows coverage and abstention against the number of calibration residues, the EM estimates of the LoF share and coverage under region and gene transfer. Table S1 gives per gene and predictor the residue-weighted coverage of each class and the abstention rate of the per-class rule and two baselines; Table S2 gives the estimated LoF shares and calibration errors behind Section 4.2. When the calibration and test halves of a gene share their LoF share, label-shift-weighted conformal prediction [28] with the true share nearly coincides with pooled conformal; with the EM estimate it inherits the estimation error, which collapsed its LoF coverage in some pairs (worst-class coverage as low as 0.039 at $\alpha = 0.1$).

S4 Strata, assay-label sensitivity and ClinVar check

For one within-gene split per gene and predictor ($\alpha = 0.1$, variant-weighted), coverage of LoF and functional variants was 0.966 and 0.929 on the third of variants farthest from the assay's class boundary, 0.929 and

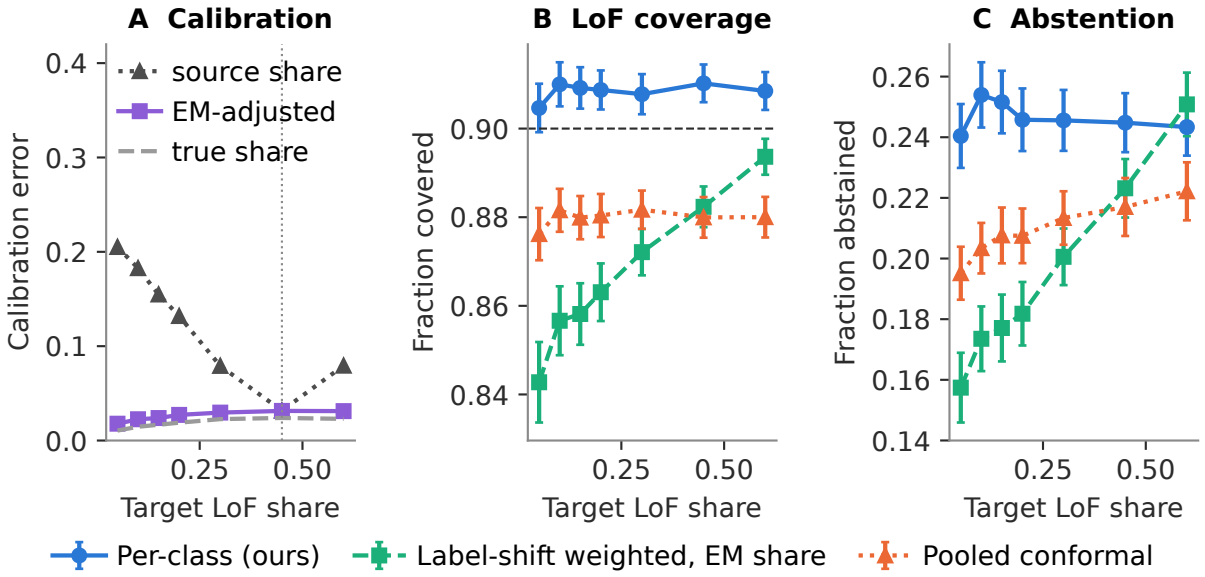


Figure S1: Simulation with residue-clustered scores, calibration at LoF share 0.45 and test share on the horizontal axis (300 replicates; error bars are 95% intervals of the mean). (A) Expected calibration error of Platt scaling at the source share, after EM prior adjustment, and after adjustment to the true share; dotted line: the calibration share 0.45. (B) Coverage of LoF variants. (C) Abstention rate.

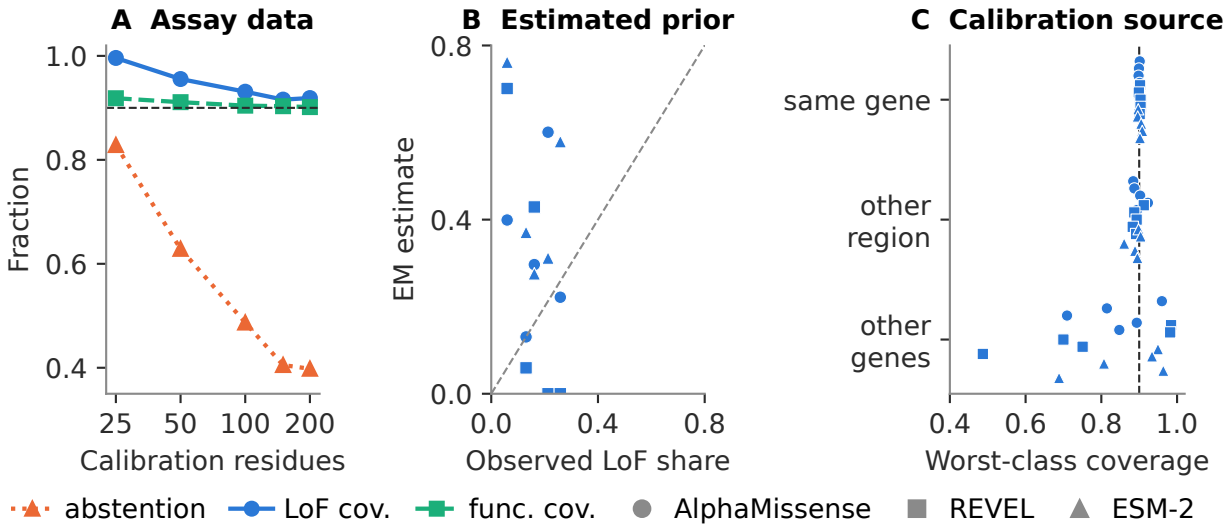


Figure S2: (A) Coverage and abstention against the number of calibration residues drawn from the gene (mean over genes, predictors and 50 draws). (B) EM estimate of the LoF share from unlabelled scores against the observed share; the dashed line is equality. (C) Worst-class coverage when calibrating on random residues of the same gene, on the other three quarters of the protein, or on the other four genes.

0.912 on the middle third, 0.865 and 0.873 on the nearest third, and 0.881 and 0.896 on variants classified VUS in ClinVar, with abstention rates of 0.334, 0.385, 0.413 and 0.392.

Table S3 gives, for each label rule, the number of labelled variants, their LoF share, the coverage and abstention of the per-class rule, and the oracle check of the certified selection (Section 4.6). This analysis is an independent run with its own random splits. Means over 50 random halvings move between runs by up to about 0.02 in coverage and 0.05 in abstention for a high-variance pair (for RAD51C with ESM-2, LoF coverage 0.895 and abstention 0.28 here against 0.915 and 0.34 in Table S1), so its main-label rows differ

Table S1: Within-gene random split, $\alpha = 0.1$, mean of 50 splits. Each cell is residue-weighted LoF coverage / functional coverage / abstention rate.

Predictor	Gene	Per-class (ours)	Pooled conformal	Fixed 0.1/0.9
AlphaMissense	BRCA1	0.916/0.901/0.25	0.952/0.889/0.32	0.971/0.927/0.44
AlphaMissense	BRCA2	0.899/0.903/0.36	0.808/0.921/0.17	0.890/0.965/0.37
AlphaMissense	BAP1	0.909/0.898/0.39	0.977/0.880/0.69	0.942/0.687/0.26
AlphaMissense	RAD51C	0.901/0.905/0.17	0.824/0.929/0.10	0.961/0.973/0.37
AlphaMissense	MSH2	0.915/0.901/0.29	0.972/0.897/0.70	0.956/0.799/0.39
REVEL	BRCA1	0.914/0.902/0.34	0.962/0.883/0.48	0.997/0.999/0.98
REVEL	BRCA2	0.912/0.899/0.42	0.841/0.911/0.24	0.947/0.998/0.68
REVEL	BAP1	0.903/0.904/0.42	0.545/0.997/0.15	0.663/1.000/0.24
REVEL	RAD51C	0.909/0.904/0.27	0.625/0.999/0.20	0.680/1.000/0.23
REVEL	MSH2	0.911/0.901/0.37	0.991/0.896/0.79	0.991/0.900/0.80
ESM-2	BRCA1	0.911/0.897/0.37	0.912/0.893/0.34	0.970/0.975/0.64
ESM-2	BRCA2	0.896/0.903/0.36	0.944/0.894/0.51	0.964/0.958/0.71
ESM-2	BAP1	0.910/0.905/0.47	0.878/0.912/0.36	0.903/0.943/0.46
ESM-2	RAD51C	0.915/0.908/0.34	0.973/0.886/0.50	0.987/0.963/0.71
ESM-2	MSH2	0.920/0.902/0.46	1.000/0.895/0.80	0.993/0.823/0.68

Table S2: Prior estimation and calibration on assayed variants outside the ClinVar labelled set. LoF share observed and estimated from unlabelled scores by expectation–maximization (EM) [26] and black-box shift estimation (BBSE) [27]; expected calibration error (10 equal-mass bins) of the ClinVar-Platt posterior, after EM prior adjustment, after adjustment to the true prior, and of Platt scaling fitted on the other four genes' assay labels. AM: AlphaMissense.

		LoF share			Expected calibration error			
Pred.	Gene	obs.	EM	BBSE	ClinVar Platt	+ EM	+ true prior	other genes
AM	BRCA1	0.161	0.297	0.316	0.203	0.135	0.064	0.049
AM	BRCA2	0.130	0.131	0.178	0.132	0.032	0.032	0.041
AM	BAP1	0.213	0.601	0.588	0.341	0.388	0.255	0.119
AM	RAD51C	0.259	0.222	0.252	0.067	0.043	0.037	0.162
AM	MSH2	0.060	0.399	0.395	0.361	0.340	0.120	0.139
REVEL	BRCA1	0.161	0.429	0.336	0.285	0.268	0.058	0.140
REVEL	BRCA2	0.130	0.059	0.149	0.176	0.071	0.025	0.111
REVEL	BAP1	0.213	0.000	0.000	0.092	0.213	0.149	0.138
REVEL	RAD51C	0.259	0.000	0.000	0.143	0.259	0.185	0.187
REVEL	MSH2	0.060	0.701	0.577	0.485	0.642	0.106	0.329
ESM-2	BRCA1	0.161	0.276	0.304	0.212	0.114	0.050	0.044
ESM-2	BRCA2	0.130	0.371	0.371	0.288	0.241	0.075	0.035
ESM-2	BAP1	0.213	0.312	0.327	0.171	0.121	0.086	0.092
ESM-2	RAD51C	0.259	0.579	0.538	0.246	0.320	0.122	0.102
ESM-2	MSH2	0.060	0.762	0.670	0.531	0.702	0.165	0.234

from Tables S1 and S8 by Monte Carlo error.

Table S4 gives the exploratory check of the calls against ClinVar labels (Section 4.4), per gene and pooled. Only BRCA1, BRCA2 and MSH2 have enough labelled residues for informative bounds; across them the LoF-call lower bounds differ by gene, so the pooled values should not be read as a common likelihood ratio. Some ClinVar classifications already use these assays or predictors as evidence, so this check is not independent clinical validation.

Table S5 counts the ClinVar missense VUS of each gene by assay status, and Table S6 gives the certified calls and mixture checks for the 147 that are unassayed but lie at assayed residues (Section 4.5).

Table S3: Sensitivity to the assay-label rule. Each gene is relabelled with the authors' classes or with the control rule at the stated posterior cut-off; other genes are unchanged. Per-class: lowest mean per-class coverage over the three predictors (50 random halvings, $\alpha = 0.1$); abstention: range over predictors; bounds held: oracle selections whose certified bound the true likelihood ratio respected (100 draws per predictor); LoF-supp.: share of draws certifying LoF-supporting calls, range over predictors. This is an independent run with its own random splits, so its main-label rows differ from Tables S1 and S8 by Monte Carlo error.

Gene	Labels	n	LoF	Per-class	Abstention	Bounds held	LoF-supp.
BRCA2	controls, 0.80	3,971	0.136	0.894	0.49–0.53	294/295	0.63–0.96
BRCA2	controls, 0.95 (main)	3,435	0.130	0.896	0.36–0.42	423/423	0.90–0.99
BRCA2	controls, 0.99	2,341	0.151	0.897	0.30–0.37	591/593	0.94–1.00
BAP1	authors (main)	4,294	0.213	0.901	0.39–0.47	460/460	0.90–1.00
BAP1	controls, 0.95	4,221	0.100	0.899	0.13–0.19	777/777	1.00
RAD51C	authors (main)	2,225	0.259	0.895	0.18–0.28	558/561	0.84–0.99
RAD51C	controls, 0.95	832	0.538	0.902	0.07–0.19	868/868	0.98–1.00

Table S4: Calls judged against ClinVar labels (exploratory). Thresholds ($\alpha = 0.1$) are set on each gene's assay labels; one ClinVar variant with $\geq$2 stars is kept per residue. n: kept P/LP and B/LB variants; calls: LoF (or functional) calls among P/LP / among B/LB variants; L^+, U^-: likelihood-ratio bounds from one-sided Clopper–Pearson endpoints at level 0.025 each (simultaneous level 0.95 per ratio). Pooled rows sum the kept counts over genes, each gene with its own thresholds. BAP1 and RAD51C have too few labelled residues to be informative.

Pred.	Gene	n P/LP	n B/LB	LoF calls	L^+	func. calls	U^-
AM	BRCA1	75	170	56/1	19.6	8/153	0.24
AM	BRCA2	60	210	49/8	9.44	6/164	0.29
AM	BAP1	6	8	5/0	0.97	1/8	1.02
AM	RAD51C	8	3	5/1	0.27	2/2	6.90
AM	MSH2	72	55	50/6	2.58	9/46	0.31
AM	pooled	221	446	165/16	11.9	26/373	0.21
REVEL	BRCA1	75	170	48/10	4.94	8/134	0.28
REVEL	BRCA2	60	210	44/9	7.56	11/182	0.37
REVEL	BAP1	6	8	4/0	0.60	2/8	1.23
REVEL	RAD51C	8	3	6/0	0.49	2/3	2.23
REVEL	MSH2	72	55	43/3	3.14	5/38	0.28
REVEL	pooled	221	446	145/22	7.99	28/365	0.23
ESM-2	BRCA1	75	170	47/1	15.7	10/159	0.26
ESM-2	BRCA2	60	210	33/4	8.66	9/184	0.32
ESM-2	BAP1	6	8	5/0	0.97	1/5	2.62
ESM-2	RAD51C	8	3	5/0	0.35	1/1	62.7
ESM-2	MSH2	72	55	44/3	3.23	3/33	0.25
ESM-2	pooled	221	446	134/8	15.4	24/382	0.19

Figure S3 shows, for the oracle design of Section 4.4, how often each selection rule claimed each level and how often the claim held.

S5 Implementation

Inputs: ClinVar `variant_summary.txt.gz` of 26 October 2026 (SHA-256 in `data/MANIFEST.txt`); MaveDB score sets `urn:mavedb:00000097-0-2` (BRCA1), `00001279-a-1` (BRCA2), `00000662-0-1` (BAP1), `0000 0673-0-1` (RAD51C) and `00000050-a-1` (MSH2, positive scores = loss of function); Ensembl VEP REST on GRCh38 (release 116) with the AlphaMissense and REVEL annotations; ESM-2 `esm2_t33_650M_UR50D`. Distance from the class boundary is the absolute difference between a variant's assay score and the midpoint

Table S5: ClinVar missense VUS by assay status. Assayed: VUS with an assay score, including intermediate and splice-region variants (Table 1 counts only those with a binary label). Check: share of 20 mixture checks rejected ($p < 0.05$) for the unassayed VUS at other residues, run where they span at least 30 residues.

			Unassayed		Check rejected,
Gene	All	Assayed	at assayed residues	elsewhere	elsewhere (%)
BRCA1	1,909	375	29	1,505	100.0
BRCA2	3,508	1,003	0	2,505	100.0
BAP1	1,189	1,173	3	13	–
RAD51C	596	570	10	16	–
MSH2	1,707	1,597	105	5	–
Total	8,909	4,718	147	4,044	

Table S6: Certified calls on the ClinVar VUS that are unassayed but lie at assayed residues (Algorithm 1, G_4, $\delta = 0.05$; mean number of calls over 20 random 40/60 calibration/certification splits of all assayed residues), and the share of mixture checks of those VUS rejected ($p < 0.05$), run where they span at least 30 residues; where the check was run, calls are averaged over the splits it accepted.

			LoF calls		Functional calls		Check
Gene	VUS	Predictor	supp.	mod.	supp.	mod.	rejected (%)
BRCA1	29	AlphaMissense	7.9	0.2	11.7	0.0	–
		REVEL	7.2	0.0	3.8	0.0	–
		ESM-2	8.0	1.0	5.0	0.0	–
BAP1	3	AlphaMissense	2.9	0.6	0.0	0.0	–
		REVEL	2.7	0.4	0.0	0.0	–
		ESM-2	2.9	0.0	0.0	0.0	–
RAD51C	10	AlphaMissense	2.8	0.8	5.7	1.2	–
		REVEL	3.9	0.5	3.4	0.0	–
		ESM-2	3.4	0.1	4.2	0.0	–
MSH2	105	AlphaMissense	31.8	5.6	60.8	4.8	40.0
		REVEL	20.0	0.0	22.9	0.0	35.0
		ESM-2	19.1	0.8	5.0	0.0	0.0

of the two class medians in its gene, divided by the distance between the medians. Expected calibration error uses ten equal-mass bins. AUROC intervals are percentiles of 1,000 bootstrap resamples of residues. The region split divides each gene's assayed residues into four contiguous blocks with equal numbers of distinct positions. ClinVar records whose aggregate germline classification is empty (somatic-only submissions) are treated as unclassified. All random draws use NumPy generators with fixed seeds (2027 plus a fixed offset per experiment). The experiments run on a laptop CPU with Python 3.13, NumPy 2.5, SciPy 1.18, scikit-learn 1.9 and pandas 3.1; ESM-2 scoring used fair-esm 2.0 and PyTorch 2.14 on an Apple-silicon GPU. Exact versions and data checksums are in the repository. The Platt map is checked to be increasing (positive slope) every time it is fitted, and every fit in the experiments passed. Certified bounds use the conventions of Section 3: $U^- = \infty$ when $\underline{\phi}_0 = 0$, an empirical ratio $x/0$ is ∞ for $x > 0$, and $0/0$ is undefined (never a claim); boundary cases are covered by unit tests (`make test`). The VUS-shift LoF shares are residue-weighted for both the calibration sample (one variant per residue) and the VUS (each residue weighted equally, then its variants).

Comparison rules (Table 3). All rules select, for each level, the grid point with the largest yield whose bound meets the level (functional calls among α at least the LoF choice of the same strength). Uncorrected Clopper–Pearson: endpoints at level $\delta/2$ for every grid point. Katz: the one-sided log interval for a ratio of two binomial proportions [24] at level $\delta/|G|$ per ratio, with 0.5 added to every cell of a ratio that has a zero count. Percentile bootstrap: 2,000 multinomial resamples of the certification residues, the 5th percentile of

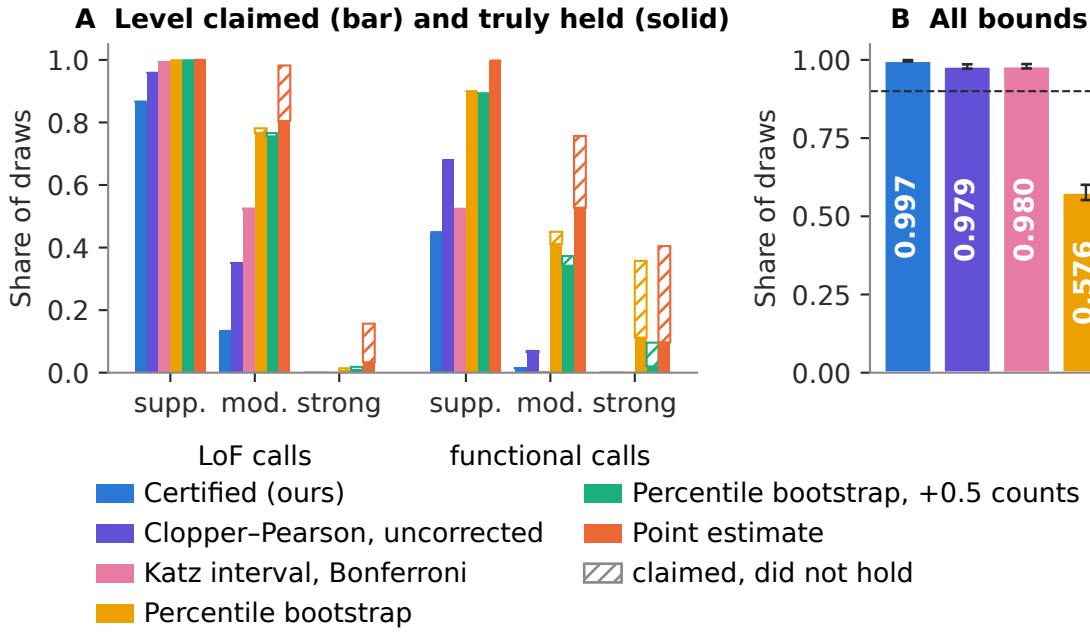


Figure S3: Threshold selection judged against the population likelihood ratio (oracle design, grid G_4, the same 1,500 draws for every rule). (A) Share of draws in which each rule claims a level (full bar) and in which it held (solid). (B) Share of draws in which every claimed bound was respected, with 95% intervals from resampling draws; a point estimate is not a bound and is omitted. Dashed line: the guaranteed 0.90.

LR^+ and the 95th of LR^-; corrected: the $100\,\delta/(2|G|)$ percentiles; +0.5 counts: 0.5 added to each of the six cells of every resample. Fixed sequence: levels tested as in Section 3, at δ/L with $L = 3$ (supporting, moderate, strong) for the simultaneous version and at δ for the per-level version.

Mixture check. The statistic of Section 3 is evaluated on the union of the observed scores, with π on 201 equally spaced values in $[0, 1]$. Each of the 300 bootstrap resamples redraws the two calibration classes with replacement at their observed sizes and a target sample of the observed size from the fitted mixture of the empirical class laws; the p-value is $(1 + \#\{D^* \geq D\})/301$. In a simulation with Gaussian classes the check rejected 4% of 100 target sets that satisfied invariance and 96% of 50 that did not. Calibration (one variant per residue) and target (one per residue) are drawn afresh in every repetition; Table S7 gives the designs.

S6 Experimental designs and all baselines

Table S7 lists the designs of the main text; every split is by residue, and calibration keeps one variant per residue unless stated. Table S8 gives, for every gene and predictor, the per-class rule next to pooled conformal prediction and the published cut-offs.

Table S7: Experimental designs.

Experiment	Residue split	Tests	Reps
Coverage (4.2)	halves: calibrate on one, test on the other	Prop. 1	50
Calibration share (4.2)	halves; calibration half subsampled to LoF share 0.05–0.5	Cor. 1	100
Reliability (4.3)	halves	Props. 1–2	200
VUS (4.3)	halves; non-VUS of one half (40% calibrate, 60% certify); VUS of the other	Cor. 1	50
Population coverage (4.3)	half of the residues drawn with replacement; coverage on all residues	Props. 1–2	200
Weighting (4.2)	halves; one per residue, all variants, residue weights	Prop. 1	50
Oracle (4.4)	30% and 50% of residues drawn with replacement; truth on all residues	Thms. 1–2, Prop. 3	100
Held-out fifth (4.4)	disjoint 30% / 50% / 20%	Thm. 1	100
Mixture check (4.5)	halves; VUS of the other half; quarters; other genes; unassayed VUS	Asm. 2	100/50/25/5/20
Use case (4.5)	all assayed residues, 40% calibrate, 60% certify; unassayed VUS	Thm. 1	20
Regions (4.6)	three contiguous quarters / the fourth	beyond Asm. 1	25/quarter

Table S8: Calls by gene and predictor ($\alpha = 0.1$; calibration and test on disjoint halves of the gene's residues, mean of 50 splits). Per-class rule: residue-weighted coverage of LoF and functional (Fun) variants and abstention rate (Abs); Monte Carlo standard errors of the coverages are 0.002–0.010. Pooled conformal: worst-class coverage and abstention. Published cut-offs (AlphaMissense classes for AM, REVEL PP3/BP4 thresholds for REVEL; none exist for ESM-2): coverage of each class and abstention. Bold: below 0.85.

		Per-class (ours)			Pooled conformal		Published cut-off		
Gene	Predictor	LoF	Fun	Abs	Worst	Abs	LoF	Fun	Abs
BRCA1	AM	0.916	0.901	0.25	0.889	0.32	0.927	**0.773**	0.13
	REVEL	0.914	0.902	0.34	0.883	0.48	1.000	**0.663**	0.59
	ESM-2	0.911	0.897	0.37	0.893	0.34	–	–	–
BRCA2	AM	0.899	0.903	0.36	**0.808**	0.17	**0.810**	0.866	0.11
	REVEL	0.912	0.899	0.42	**0.841**	0.24	0.984	**0.803**	0.57
	ESM-2	0.896	0.903	0.36	0.894	0.51	–	–	–
BAP1	AM	0.909	0.898	0.39	0.880	0.69	0.921	**0.540**	0.07
	REVEL	0.903	0.904	0.42	**0.545**	0.15	**0.799**	0.975	0.29
	ESM-2	0.910	0.905	0.47	0.878	0.36	–	–	–
RAD51C	AM	0.901	0.905	0.17	**0.824**	0.10	0.860	0.902	0.11
	REVEL	0.909	0.904	0.27	**0.625**	0.20	**0.792**	0.979	0.25
	ESM-2	0.915	0.908	0.34	0.886	0.50	–	–	–
MSH2	AM	0.915	0.901	0.29	0.897	0.70	0.951	**0.642**	0.12
	REVEL	0.911	0.901	0.37	0.896	0.79	0.995	**0.445**	0.41
	ESM-2	0.920	0.902	0.46	0.895	0.80	–	–	–